TXELECTRUM

Bitcoin Wallet

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Project Description

What is a Wallet?

Asymmetric key pair

A wallet is essentially a pair of private key and public key

Pay to a public key

People can pay to you by sharing your public key

Spend coins using Digital Signature

Digital Signature is a proof that you are the private key owner, (the wallet owner.)

Wallet Program

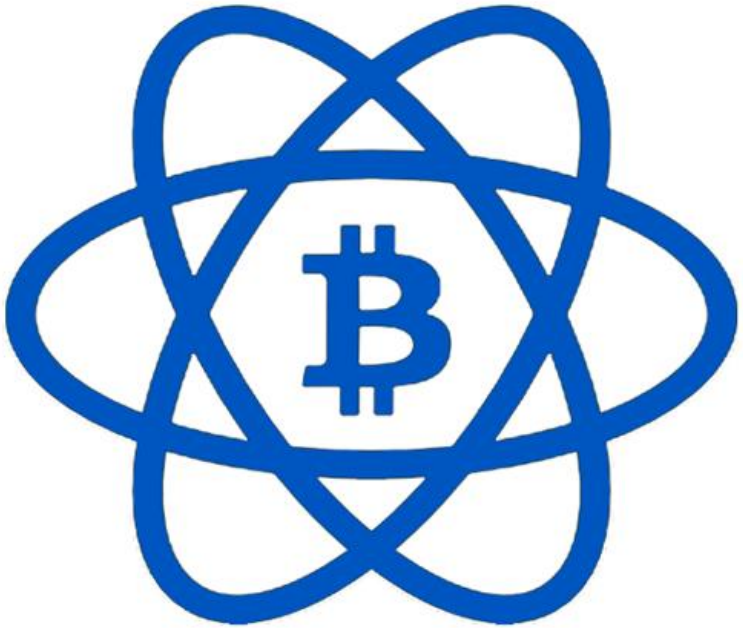
01 Creates Wallets

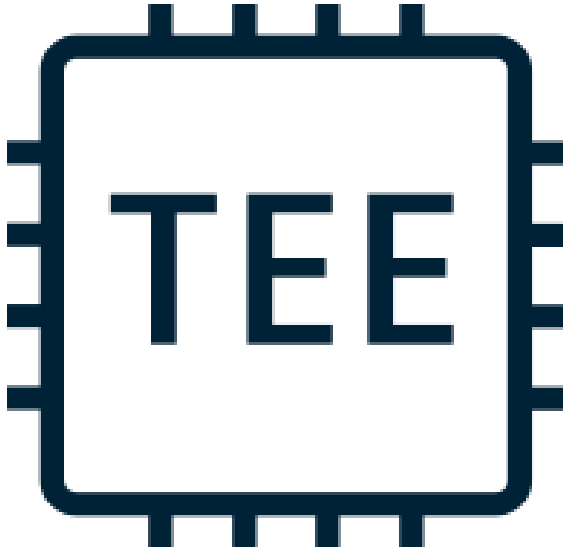
02 Creates Transactions

03 Broadcasts Transactions

02 Shows Transactions History

Electrum TEE Plugin





Wallets Theft

Wallets Theft

- If a thief stole the private key, he could use the bitcoins has he wants.
- A user must severely protect his private key.

Wallets Protections

On The Computer

- The wallet program saves the private key encrypted in the hard drive.
- A thief can steal the private key when it is in the wallet program’s memory unencrypted.

Cold Storage

- The wallet program that is connected to the Internet can only create transactions without singing (it doesn’t have the private key).
- The wallet program on an offline computer have the private key and can sign the transaction but cannot broadcast it.
- You need two computers to create this setup and it is inconvenient to sign a transaction.

Hardware Wallet

- Hardware wallet is a TEE on a USB.
- The wallet program doesn’t have access to the private key and doesn’t sign a transaction.
- The hardware wallet saves the private key and signs the transaction.
- Hardware wallet is quite expensive, and you need to hold it whenever you want to create a transaction.

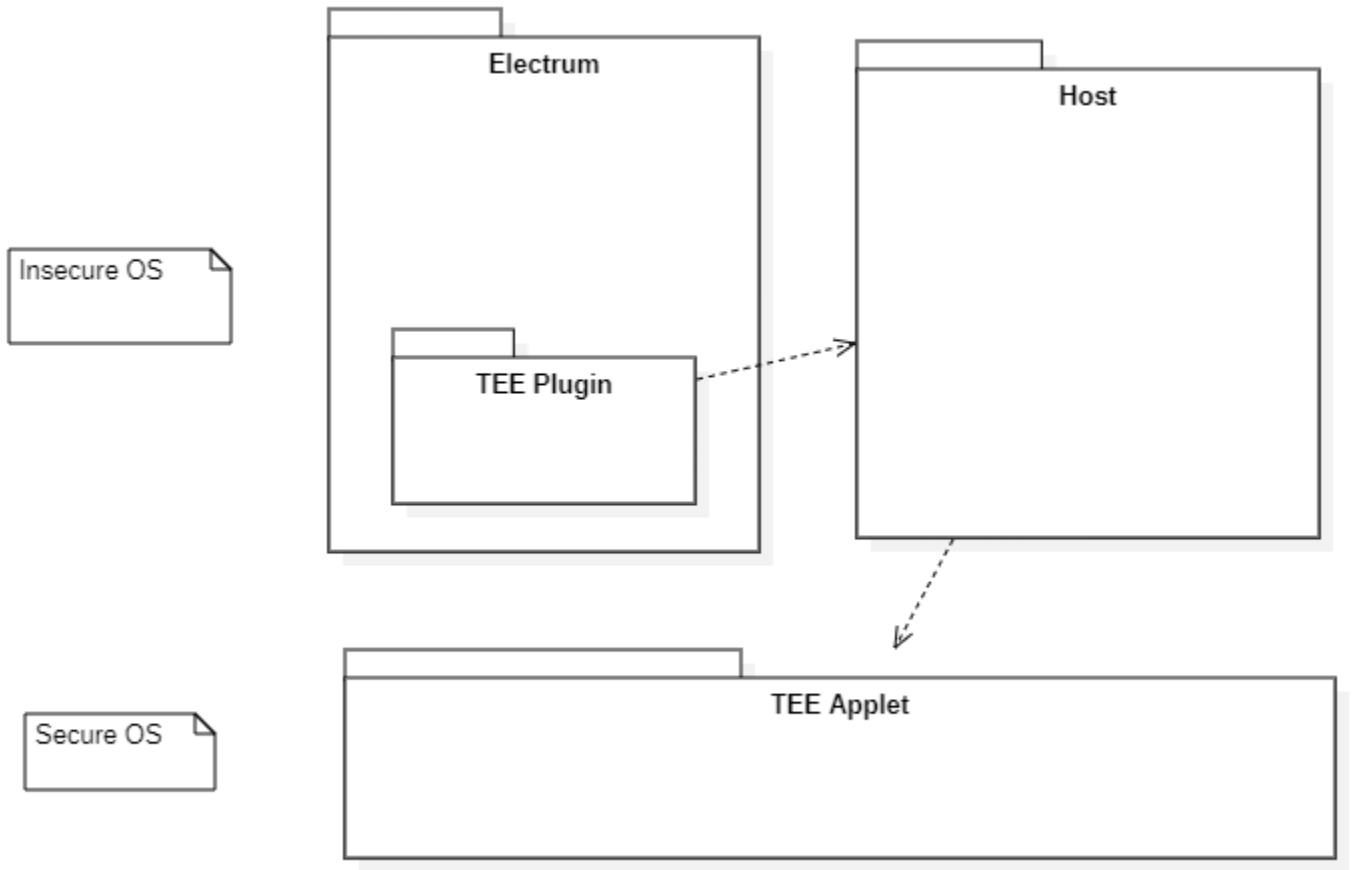
Our Solution

Our Solution

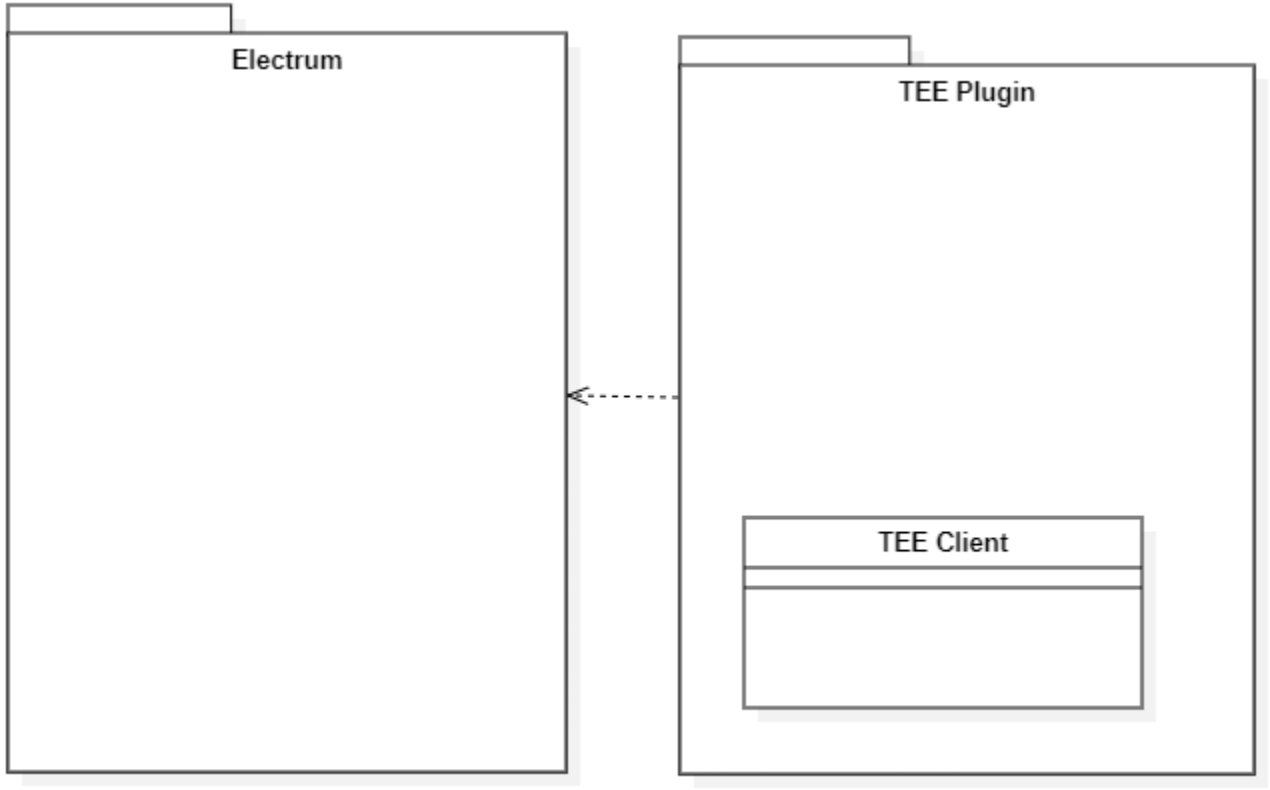
- Use the TEE that comes with every Intel chipset.
- The TEE holds the private key and signs the transactions.
- It’s secure and it’s as convenient as saving the private key on the hard drive.

Architecture

High-Level View



Electrum Plugin



REST API (OpenApi)

Create key pair

POST http://localhost:51841/create_keypair

Request

Body

hashed_password string<binary>

SHA-1 hash binary

>= 40 characters <= 40 characters

Example: 9f447054819975f30a80e593961d850788edf9b4

Match pattern: ^[A-Fa-f0-9]{40}$

Responses

200 400 500

OK

Body

public_key string<binary>

Representation of secp256k1 public key in SEC's compressed format.

>= 66 characters <= 66 characters

Example: 02f2dd109034c408433042ab0d237436ec62b5475ad87121ab607503c7d67d0895

Match pattern: ^[A-Fa-f0-9]{66}$

Sign a buffer

POST http://localhost:51841/sign_buffer

Request

Body

public_key string<binary>

Representation of secp256k1 public key in SEC's compressed format.

>= 66 characters <= 66 characters

Example: 02f2dd109034c408433042ab0d237436ec62b5475ad87121ab607503c7d67d0895

Match pattern: ^[A-Fa-f0-9]{66}$

hashed_password string<binary>

SHA-1 hash binary

>= 40 characters <= 40 characters

Example: 9f447054819975f30a80e593961d850788edf9b4

Match pattern: ^[A-Fa-f0-9]{40}$

hashed_buffer string<binary>

SHA-256 hash binary

>= 64 characters <= 64 characters

Example: 315f5bdb76d078c43b8ac0064e4a0164612b1fce77c869345bfc94c75894edd3

Match pattern: ^[A-Fa-f0-9]{64}$

Responses

200 400 401 404 500

OK

Body

signature string<binary>

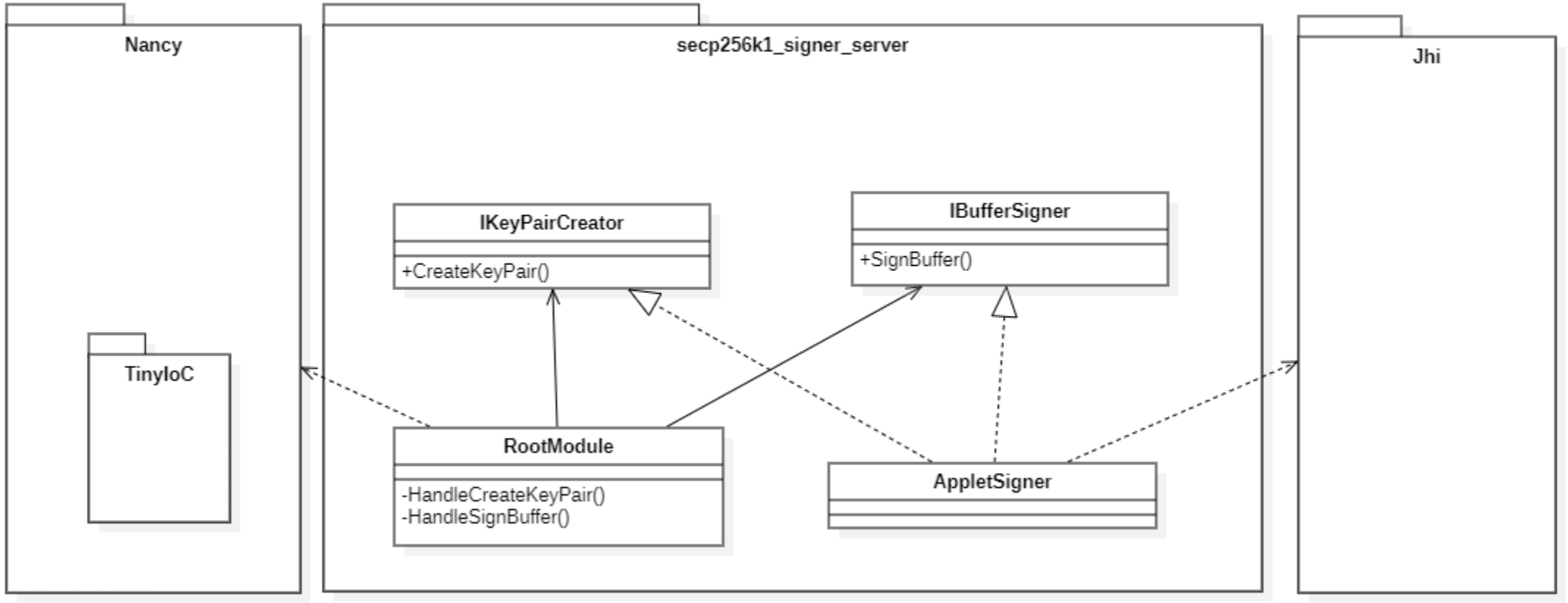
Representation of secp256k1 digital signature in DER format.

Example: 3045022100f8a4dbab9c9fed17dfe75db206c2c070f86ae8537b87a221906bf6c981dd1259
6aab19c0efda4c0a965b1a6ceb5

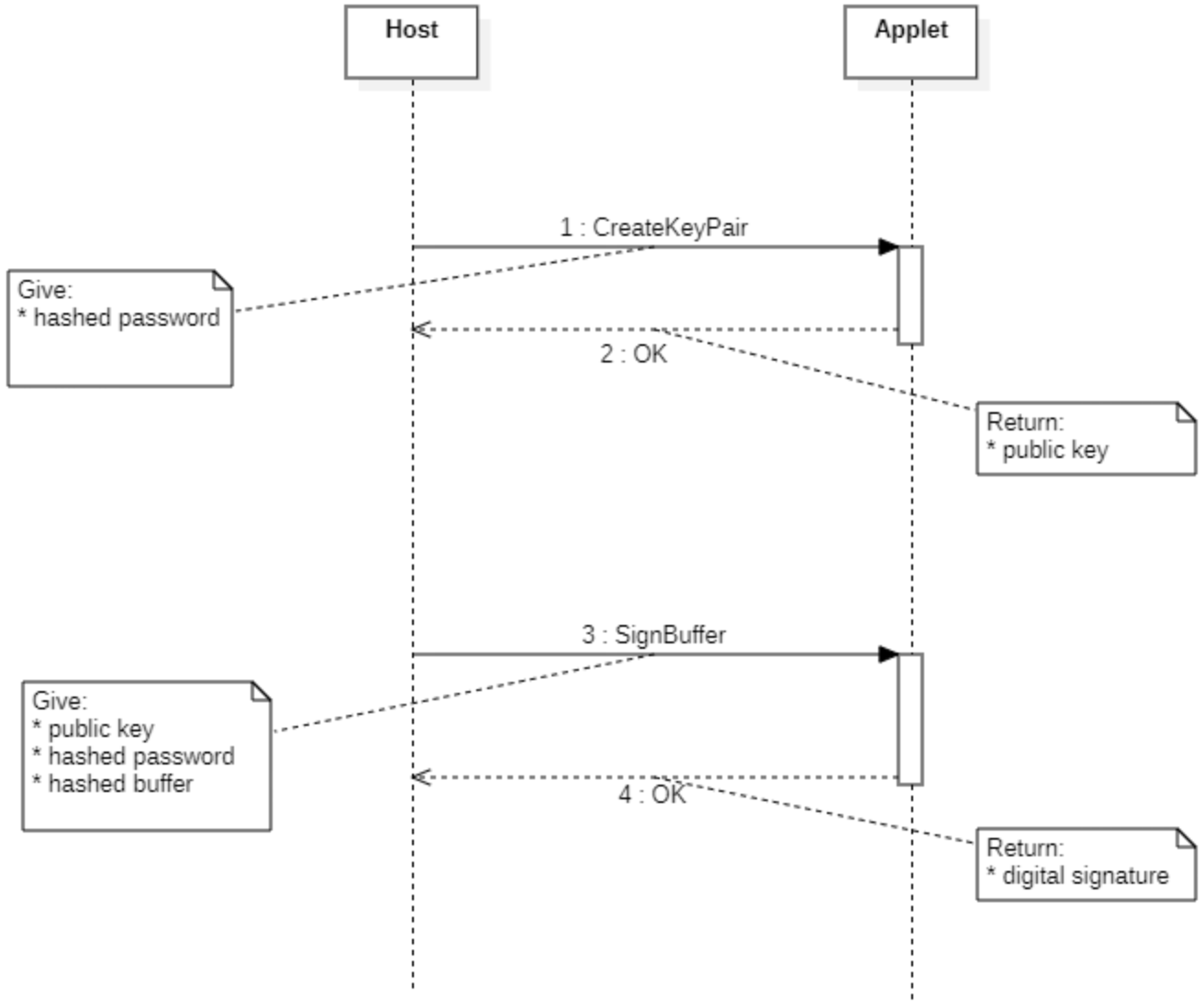
Match pattern: ^[A-Fa-f0-9]*$

https://yonieuler.stoplight.io/docs/secp256k1-signer/ou7gl61whyceb-secp-256-k1-signer

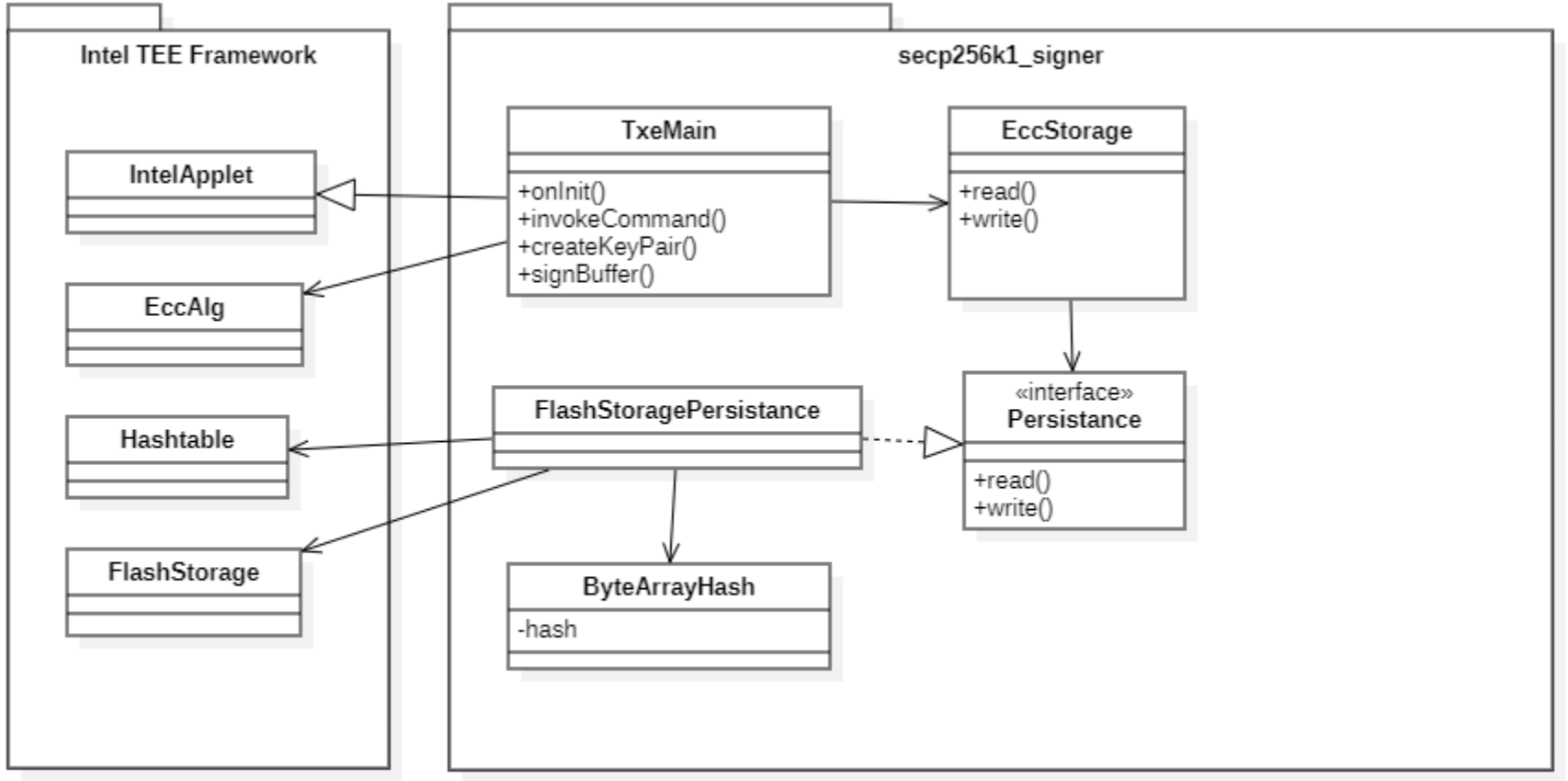
C# REST Server & Host



Host-Applet Protocol



TEE Applet


Intel TEE Framework
IntelApplet
EccAlg
Hashtable
FlashStorage
secp256k1_signer
TxeMain
+onInit()
+invokeCommand()
+createKeyPair()
+signBuffer()
EccStorage
+read()
+write()
FlashStoragePersistance
«interface»
Persistance
+read()
+write()
ByteArrayHash
-hash

Threat-Model

Threat-Model

Bad	Good
When creating new wallet, the thief can impersonate the TXE server and steal the private key.	If a user successfully created a wallet on the TXE, the private key never comes out.
The http connection is not secure, and the thief can steal the password and the public key.	The thief can create transactions only when he has access to the machine.
The attacker can DoS the TXE by creating many wallets.	

Future Work

Future Work

Features	Security
Delete wallets.	Create secure http connection between the client and the server using SIGMA protocol.
Create hierarchy wallets (BIP32).	Securely create or enter a password using secure-display.
Export the private key using a seed (BIP39).	Securely export the private key's seed using secure-display.
Other latest features of Bitcoin Blockchain.	

Libraries

Libraries

Python	C#
Electrum’s libraries	Nancy (REST server)
requesets (REST client)	TinyIoC (Inversion of Control library)
hashlib (hash library)	Jhi (TEE Communication)

Challenges

Challenges

- Understand formats that Bitcoin uses.
  - Public key format (SEC)
  - Bitcoin address format
  - Digital Signature format (DER)
  - Transaction format

- Read Electrum’s source code.

- Use Intel’s small java libraries.

Demo

Thank you

Do you have any questions?